Rathod Sai Dhanush

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.96
SRI CHAITANYA JNR KALASALA , MADHAPUR,HYDERABAD,TELANGANA	Class XII (Telangana State Board of Intermediate Education)	2023	9.43%
SRI CHAITANYA HIGH SCHOOL,KUKATPALLY, MEDCHAL DISTRICT	Class X (Telangana State Board of Secondary Education)	2021	10.00%

SKILLS

Programming Languages : Programming: Python, C++, C, SQL

Frameworks & Libraries : PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, Flask, NumPy, Pandas, Matplotlib

Developer Tools : Git, GitHub, Google Colab, Jupyter, VS Code, LaTeX.

Technologies : REST APIs, Pixhawk, ArduPilot, Fusion 360, Solid Edge, 3D Printing

RELEVANT COURSES

Mathematics : Numerical Analysis, Introduction to Bayesian Statistics, Calculus, Differential Equations, Series of Functions, Introduction to Metric Spaces, Matrix Theory, Applied Statistics, Probability and Random Variables.

Core Courses : Computer Vision, Natural Language Processing, Deep Learning, Reinforcement Learning, Foundations of Machine Learning, Convex Optimization,Data Science Analysis, Robotics.

Computer Science : Introduction to Programming, Discrete Mathematics, Data Structures, Computer Architecture, Operating Systems, Database Management Systems, Compilers.

PROJECTS

Embedded Systems Project - LED Blinking using RISC-V Assembly

- Programmed a **GPIO-controlled LED blinking loop** in **RISC-V assembly language**, directly manipulating memory-mapped registers for pin configuration and timing delays.
- Used simple delay loops and GPIO register access to turn an LED on and off, demonstrating basic **embedded system control**.

Unmanned Surface Vehicle (USV) Design & Mission Planning | Fusion 360, Pixhawk, ArduPilot

- Designed and fabricated a **1.22 × 1.32 m autonomous USV**, developing the CAD model in **Fusion 360** and fabricating a PETG-HS 3D-printed hull with watertight joints.
- Integrated **Pixhawk X7 Pro+, GPS, dual T200 thrusters, ESCs, and RC modules** for autonomous vehicle control and operation.
- Implemented **GPS waypoint navigation** using ArduPilot Mission Planner and analyzed **drag-thrust characteristics**, estimating a **2 m/s maximum speed**.

Counterfactual Explanations for CNN-Based Pet Breed Classification :

- Fine-tuned **ImageNet-pretrained ResNet18** for **37-class pet-breed classification**, achieving **87.49% test accuracy**

on Oxford-IIIT Pets.
- Developed a **gradient-based counterfactual pipeline** using target, L1, total-variation and blur regularization to generate minimal prediction-changing perturbations.
- Analyzed **difference and overlay maps** to identify discriminative features, highlighting **eyes, muzzle, ears and coat texture**.

Cultural-Specific Named Entity Recognition | Python, BERT, Hugging Face :

- Created a **12K-sample cultural NER dataset** with **15+ Indian cultural entity categories** using template-based data augmentation and BIO tagging.
- Fine-tuned **BERT for token classification**, achieving an **F1-score of 86.49%** on unseen test data.
- Improved recognition of **multi-word and underrepresented cultural entities** across food, festivals, rituals, and traditions through balanced augmentation.

Bayesian Inference & Model Comparison | Python, MCMC, Gibbs Sampling, Metropolis-Hastings

- Built a **Bayesian linear regression model** on **3,000 player observations** to analyze relationships between player attributes and overall ratings.
- Implemented **Gibbs Sampling and Metropolis-Hastings** for posterior estimation and compared parameter convergence across MCMC methods.
- Applied **Bayes Factor analysis and Posterior Predictive Checks** to compare competing models and evaluate predictive adequacy.

POSITIONS OF RESPONSIBILITY

MILAN PRONITES:

- Coordinated Pronites events at **Milan Fest**, managing logistics and guest interactions.
- Ensured smooth execution of professional talks by handling scheduling and on-ground operation.

Events Head - Diesta (Institute Departmental Sports Fest)

- Led end-to-end planning and execution of the institute-level sports fest, coordinating over **10+ sports events**.
- Supervised a team of volunteers for logistics, registrations, and on-ground management.

Sponsorship Coordinator - MILAN

- Coordinating sponsorship activities for MILAN Fest, focusing on sponsor outreach and partnership development.
- Supporting sponsorship management and ensuring effective coordination with sponsors for successful event execution.

Sponsorship Coordinator - Abhyant

- Coordinated sponsorship activities for Abhyant 2026, contributing to sponsor outreach and partnership development.
- Supported sponsorship management and coordination to ensure smooth execution of the fest.

Volleyball Coordinator - (2025-26)

- Coordinated the institute's volleyball team, managing team activities and event organization.
- Demonstrated strong leadership, organizational, and communication skills to ensure effective team coordination and successful event execution.

EXTRACURRICULAR

Inter IIT Sports Meet - Volleyball Team Member :

- Representing IIT Hyderabad in Volleyball at the prestigious **Inter IIT Sports Meet 2025**, competing against top institutes across the country.
- Participated in regular training sessions and matches as part of the official institute team.

EXPERIENCE

Deccan AI -- AI/ML Intern

- Evaluated **LLM-generated responses** across diverse tasks, comparing outputs against defined criteria for correctness, relevance, reasoning, and instruction adherence.
- Identified model failure cases and provided **structured corrections, feedback, and improved responses**,

contributing high-quality evaluation data for AI system improvement.
- Worked on **AI model evaluation and data-quality workflows**, systematically analyzing model behavior and iterating on task outputs to improve reliability and performance.